Matrix Theory I: Special Matrices, Trace, Determinants, Adjoint and Rank

From the class book: Chapter 1, Sections 1.1 to 1.5

1. Key definitions

- $M_{m\times n}(F)$ is the set of $m \times n$ matrices over a field F, $M_n(F) = M_{n\times n}(F)$.
- A^T is the transpose, $A^* = \bar{A}^T$ the conjugate transpose, I_n the identity and J_n the $n \times n$ matrix with every entry 1.

Special square matrices. A is **symmetric** if $A^T = A$, **skew-symmetric** if $A^T = -A$, **Hermitian** if $A^* = A$, **skew-Hermitian** if $A^* = -A$, **orthogonal** if $A^T A = I$, **unitary** if $A^* A = I$, **normal** if $AA^* = A^* A$, **idempotent** if $A^2 = A$, **involutory** if $A^2 = I$, and **nilpotent** if $A^k = 0$ for some $k \geq 1$ (the least such k is the index).

Type	Structure	Determinant	Eigenvalues
Real symmetric	$a_{ij} = a_{ji}$	real	real
Real skew-symmetric	$a_{ii} = 0,\ a_{ij} = -a_{ji}$	0 if n odd; ≥ 0 if n even	0 or purely imaginary
Hermitian	$a_{ii} \in \mathbb{R},\ a_{ij} = \overline{a_{ji}}$	real	real
Skew-Hermitian	$a_{ii} \in i\mathbb{R}$	0 or purely imaginary (n odd), real (n even)	0 or purely imaginary
Orthogonal / unitary	orthonormal columns	± 1 / $\lvert\det\rvert = 1$	$\lvert\lambda\rvert = 1$
Idempotent	$A^2 = A$	0 or 1	$0, 1$; $\operatorname{tr} A = \operatorname{rank} A$
Involutory	$A^2 = I$	± 1	± 1
Nilpotent	$A^k = 0$	0	all 0; $\operatorname{tr} A = 0$
Permutation	one 1 in each row and column	sign of the permutation	roots of unity

Important results: decompositions and dimensions

- Every square matrix is uniquely symmetric + skew-symmetric: $A = \frac{1}{2}(A + A^T) + \frac{1}{2}(A - A^T)$.
- Every complex matrix is $H + iK$ with H, K Hermitian.
- As real vector spaces, symmetric $n \times n$ matrices have dimension $\frac{n(n+1)}{2}$, skew-symmetric $\frac{n(n-1)}{2}$, Hermitian and skew-Hermitian n^2, upper triangular $\frac{n(n+1)}{2}$, trace-zero $n^2 - 1$.
- If A is idempotent, so is $I - A$, $\operatorname{rank} A + \operatorname{rank}(I - A) = n$, and $2A - I$ is involutory.

2. Main results

Theorem (Trace)

- tr is linear, $\operatorname{tr} A^T = \operatorname{tr} A$, $\operatorname{tr}(AB) = \operatorname{tr}(BA)$ and $\operatorname{tr}(ABC) = \operatorname{tr}(BCA) = \operatorname{tr}(CAB)$ (cyclic only).
- $\operatorname{tr}(P^{-1}AP) = \operatorname{tr} A$.
- $\operatorname{tr}(A^*A) = \sum |a_{ij}|^2$, so $\operatorname{tr}(A^*A) = 0 \iff A = 0$.
- In characteristic 0 there are no A, B with $AB - BA = I$.

Theorem (Determinants)

- $\det A^T = \det A$, $\det(AB) = \det A \det B$, $\det(cA) = c^n \det A$.
- A row swap changes the sign, adding a multiple of a row changes nothing.
- Triangular: product of the diagonal.
- Block: $\det \begin{pmatrix} A & B \\ 0 & D \end{pmatrix} = \det A \det D$, and $\det \begin{pmatrix} A & B \\ C & D \end{pmatrix} = \det A \, \det(D - CA^{-1}B)$ if A is invertible.
- $\det A \neq 0 \iff A$ invertible; $\det A$ is the product of the eigenvalues.

Theorem (Adjoint)

- $A \operatorname{adj} A = (\operatorname{adj} A)A = (\det A)I$, so $A^{-1} = \operatorname{adj} A / \det A$.
- For $n \geq 2$: $\det(\operatorname{adj} A) = (\det A)^{n-1}$, $\operatorname{adj}(\operatorname{adj} A) = (\det A)^{n-2} A$, $\operatorname{adj}(cA) = c^{n-1} \operatorname{adj} A$, $\operatorname{adj}(AB) = \operatorname{adj} B \operatorname{adj} A$.
- $\operatorname{rank}(\operatorname{adj} A) = n, 1, 0$ according as $\operatorname{rank} A = n,\ n-1,\ \leq n-2$.

Theorem (Rank inequalities)

For A ($m \times n$) and B ($n \times p$): $\operatorname{rank}(AB) \le \min(\operatorname{rank} A, \operatorname{rank} B)$; Sylvester: $\operatorname{rank} A + \operatorname{rank} B - n \le \operatorname{rank}(AB)$; $\operatorname{rank}(A+B) \le \operatorname{rank} A + \operatorname{rank} B$; $\operatorname{rank}(PAQ) = \operatorname{rank} A$ for invertible P, Q; over $\mathbb{R}$, $\operatorname{rank}(A^T A) = \operatorname{rank}(AA^T) = \operatorname{rank} A$ (over $\mathbb{C}$ use $A^* A$); Frobenius: $\operatorname{rank}(AB) + \operatorname{rank}(BC) \le \operatorname{rank} B + \operatorname{rank}(ABC)$.

Rank normal form.

- If $\operatorname{rank} A = r$ there are invertible P, Q with $PAQ = \begin{pmatrix} I_r & 0 \\ 0 & 0 \end{pmatrix}$; two $m \times n$ matrices are equivalent iff they have the same rank.
- Row operations keep the row space, the rank and the linear relations among the columns; two matrices have the same row space iff they have the same RREF.

3. Standard values

Matrix	Value
$\det(aI_n + bJ_n)$	$(a+nb)\,a^{n-1}$; x on the diagonal, y elsewhere: $(x+(n-1)y)(x-y)^{n-1}$
$\det(J_n - I_n)$	$(-1)^{n-1}(n-1)$
$\det(I + uv^T)$, $\det(A+uv^T)$	$1 + v^T u$, $\det A\,(1 + v^T A^{-1} u)$
Vandermonde $\det[x_i^{j-1}]$	$\prod_{i<j}(x_j - x_i)$
Anti-diagonal permutation matrix	$(-1)^{n(n-1)/2} = (-1)^{\lfloor n/2 \rfloor}$
4×4 skew-symmetric $[a,b,c;d,e;f]$	$(af - be + cd)^2$ (Pfaffian squared)
3×3 circulant (a,b,c)	$a^3 + b^3 + c^3 - 3abc$; the matrix $\begin{pmatrix} a & b & c \\ b & c & a \\ c & a & b \end{pmatrix}$ has the negative of this
Tridiagonal $2, -1$	$n+1$
$[\min(i,j)]$	1
Rank of $[i+j]$, $[ij]$, $[(i-j)^2]$, $[\sin(i+j)]$	$2,\ 1,\ 3,\ 2$ (for $n \ge 3$)
Matrices of rank m in $M_{m\times n}(\mathbb{F}_q)$	$(q^n-1)(q^n-q)\cdots(q^n-q^{m-1})$; $\lvert GL_2(\mathbb{F}_3)\rvert = 48$, $\lvert GL_3(\mathbb{F}_2)\rvert = 168$

Method: a patterned determinant.

- (1) If every row sums to s, add all columns to the first and take s out.
- (2) Subtract the first row from the others; $aI + bJ$ and "diagonal plus constant" become triangular.
- (3) If two rows become proportional at some parameter value, that value is a root.
- (4) Check the answer at $n = 2$ or $x = 0$ before marking it.

Method: rank of a numerical matrix.

- Remove obvious dependencies first (equal, proportional or sum rows), then row reduce and count non-zero rows.
- A non-zero determinant gives full rank at once.
- With a parameter, the rank drops only where the leading minors vanish.

4. Exam shortcuts and common traps

- For 2×2: $A^2 = (\operatorname{tr} A)A - (\det A)I$ and $\operatorname{tr} A^2 = (\operatorname{tr} A)^2 - 2\det A$. Also $\det(A+B) + \det(A-B) = 2\det A + 2\det B$.
- Row sums all equal to c: $(1, \dots, 1)^T$ is an eigenvector for c, and the sum of all entries of A^k is nc^k.
- $\operatorname{rank} A = 1 \iff A = uv^T$ $(u, v \neq 0)$; then $A^2 = (\operatorname{tr} A)A$: nilpotent if $\operatorname{tr} A = 0$, diagonalizable otherwise.
- $AB = 0$ with A of size $m \times n$ gives $\operatorname{rank} A + \operatorname{rank} B \leq n$; so $A^2 = 0$ in M_n forces $\operatorname{rank} A \leq n/2$.
- $AB = I_m$ (A is $m \times n$) forces $\operatorname{rank} A = \operatorname{rank} B = m \leq n$, and $BA \neq I_n$ when $m < n$.
- An integer matrix has an integer inverse iff $\det A = \pm 1$. If N is nilpotent, $I + N$ is invertible.
- **Traps:** $\det(A+B) \neq \det A + \det B$; $\det(cA) = c^n \det A$, not $c \det A$; $\operatorname{tr}(ABC) \neq \operatorname{tr}(BAC)$ in general; idempotent need not be symmetric; $A^2 = 0$ need not give $A = 0$; $\operatorname{rank}(A^TA) = \operatorname{rank} A$ fails over $\mathbb{C}$ ($A = (1, i)^T$) and over finite fields; a skew-symmetric matrix has even rank, so a 3×3 real skew-symmetric matrix is singular.

5. Solved examples

Example 1. Evaluate $\det \begin{pmatrix} 1+x & 1 & 1 & 1 \\ 1 & 1-x & 1 & 1 \\ 1 & 1 & 1+y & 1 \\ 1 & 1 & 1 & 1-y \end{pmatrix}$.

Solution

- Subtract row 4 from rows 1, 2, 3: they become $(x,0,0,y)$, $(0,-x,0,y)$, $(0,0,y,y)$, and row 4 stays $(1,1,1,1-y)$.
- For $x, y \neq 0$ apply $C_4 \to C_4 - \frac{y}{x}C_1 + \frac{y}{x}C_2 - C_3$: the first three rows get 0 in column 4 and row 4 becomes $(1,1,1,-y)$.
- The matrix is now lower triangular with diagonal $x, -x, y, -y$, so $\det = x^2y^2$ (and by continuity also when x or y is 0).
- Check: it vanishes at $x = 0$ (rows 1, 2 equal).

Example 2. For $A = \begin{pmatrix} 5 & 9 & 8 \\ 1 & 8 & 2 \\ 9 & 1 & 0 \end{pmatrix}$, decide whether A^{-1} has integer entries and whether $13 \mid \det A$.

Solution

- $\det A = 5(0-2) - 9(0-18) + 8(1-72) = -416 = -2^5 \cdot 13$.
- Since $\det A \neq \pm 1$, A^{-1} is not an integer matrix.
- Without expanding: $598, 182, 910$ are multiples of 13, so $C_3 \to C_3 + 100C_1 + 10C_2$ makes the third column divisible by 13.

Example 3. Find $\det A$ for the 4×4 matrix with 0 on the diagonal and 1 elsewhere, and for the 6×6 matrix with 1 on the diagonal, 2 on the anti-diagonal and 0 elsewhere.

Solution

- The first is $J_4 - I_4$: $\det = (-1)^3 \cdot 3 = -3$.
- For the second, reorder rows and columns in the pairs $\{1,6\}, \{2,5\}, \{3,4\}$ (the same permutation on both, so no sign change): three blocks $\begin{pmatrix} 1 & 2 \\ 2 & 1 \end{pmatrix}$ of determinant -3, so $\det = (-3)^3 = -27$.

Example 4. Let $A \in M_3(\mathbb{R})$, $\det A = 4$. Find $\det(\operatorname{adj}(2A))$, $\det(2 \operatorname{adj} A)$ and $\operatorname{adj}(\operatorname{adj} A)$.

Solution

- $\det(2A) = 8 \cdot 4 = 32$, so $\det \operatorname{adj}(2A) = 32^2 = 1024$.
- $\det(2 \operatorname{adj} A) = 2^3 (\det A)^2 = 128$.
- $\operatorname{adj}(\operatorname{adj} A) = (\det A)^{3-2} A = 4A$.

Example 5. Find the rank of $\begin{pmatrix} 0 & 1 & 0 & 1 \\ -1 & 0 & 1 & 0 \\ 0 & -1 & 0 & 1 \\ -1 & 0 & -1 & 0 \end{pmatrix}$, and of the 4×4 matrix $[i + j]$.

Solution

- The first is skew-symmetric with $a = 1, b = 0, c = 1, d = 1, e = 0, f = 1$: Pfaffian $af - be + cd = 1 - 0 + 1 = 2 \neq 0$, so $\det = 4$ and the rank is 4.
- For $[i + j]$: every row is $i(1, 1, 1, 1) + (1, 2, 3, 4)$, so the rank is at most 2, and rows 1 and 2 are independent. Rank 2.

Example 6. Let A be 4×7 and B be 7×4 with $AB = I_4$. Which hold: $\operatorname{rank} A = 4$; $\operatorname{rank} B = 7$; $\operatorname{nullity} B = 0$; $BA = I_7$?

Solution

- $4 = \operatorname{rank}(AB) \leq \operatorname{rank} A, \operatorname{rank} B \leq 4$, so both ranks are 4.
- Then $\operatorname{nullity} B = 4 - 4 = 0$ (B acts on $\mathbb{R}^4$).
- $\operatorname{rank} B = 7$ is impossible, and BA has rank $4 < 7$, so $BA \neq I_7$. True: the first and third.

Example 7. Find the number of 3×3 matrices over $\mathbb{F}_2$ of rank 3, and the number of 2×2 matrices over $\mathbb{F}_3$ with determinant 1.

Solution

- Choose independent rows one at a time: $(8 - 1)(8 - 2)(8 - 4) = 168$.
- Over $\mathbb{F}_3$, $|GL_2(\mathbb{F}_3)| = (9 - 1)(9 - 3) = 48$; the determinant map onto $\mathbb{F}_3^\times = \{1, 2\}$ has equal fibres, so $48/2 = 24$ have determinant 1.